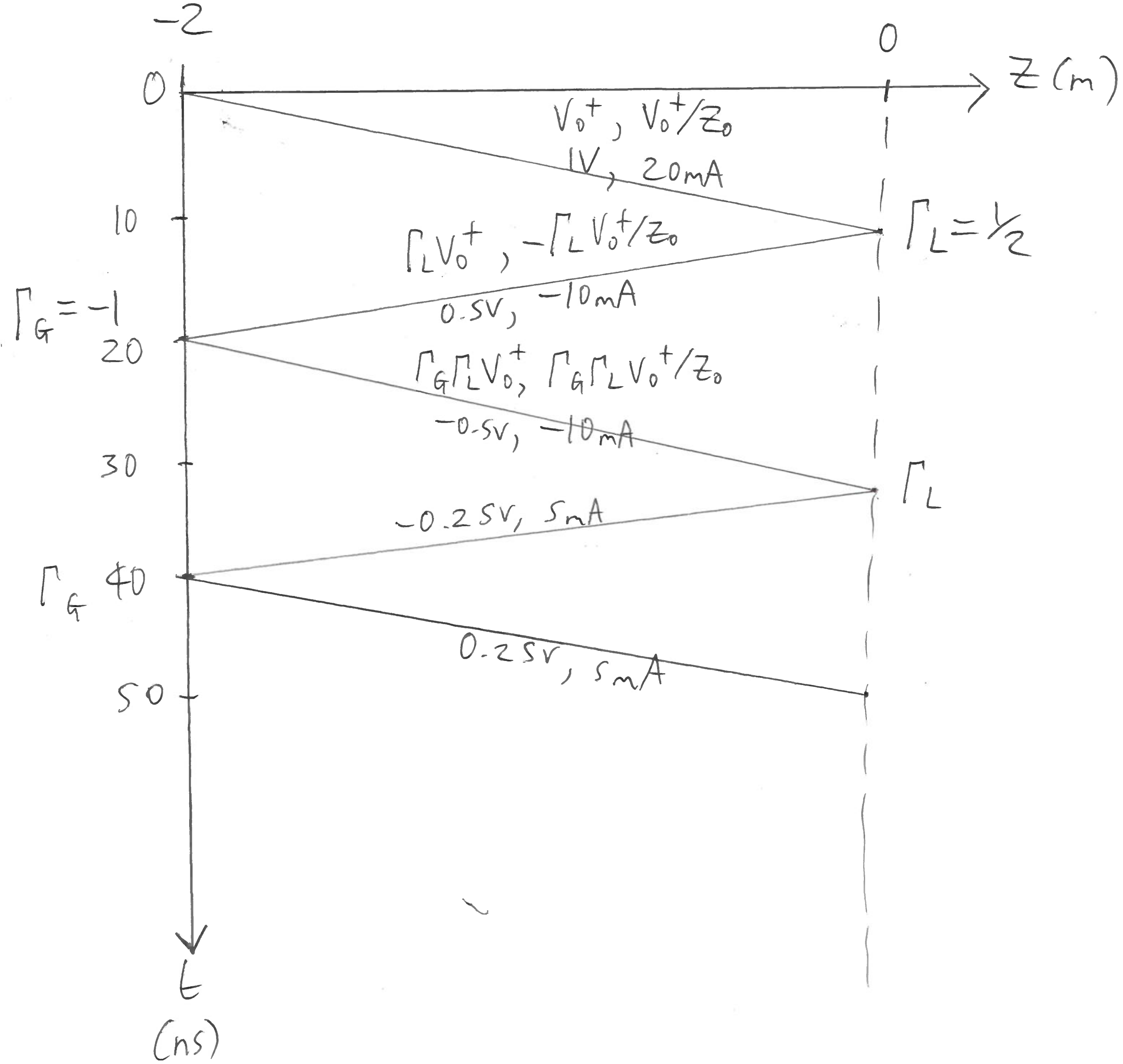
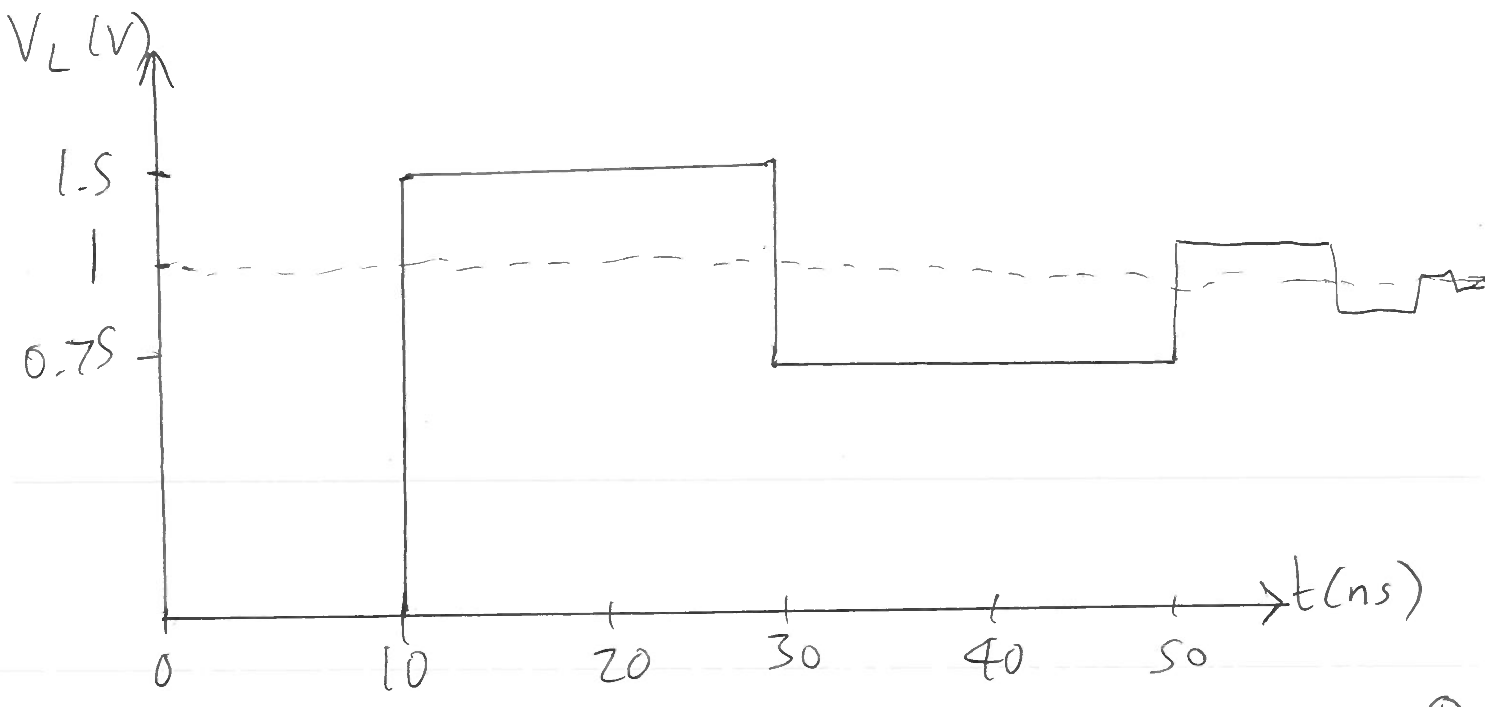
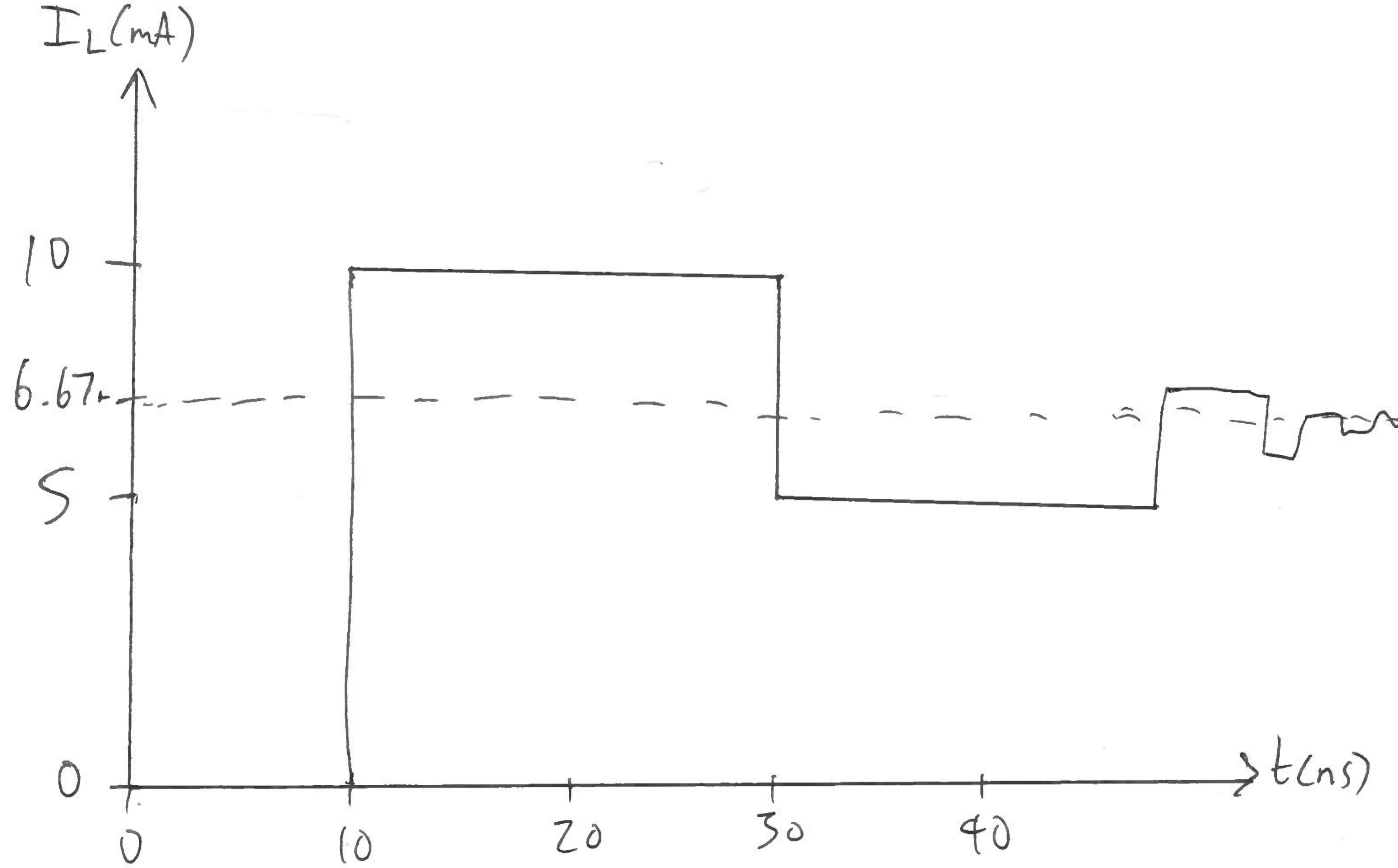


-2
0
0
z (m)
V_0^+, V_0^+/Z_0
1V, 20mA
10
$\Gamma_L = 1/2$
$\Gamma_L V_0^+$, $-\Gamma_L V_0^+/Z_0$
0.5V, -10mA
$\Gamma_G = -1$
20
$\Gamma_G \Gamma_L V_0^+$, $\Gamma_G \Gamma_L V_0^+/Z_0$
-0.5V, -10mA
30
Γ_L
-0.25V, 5mA
Γ_G
40
0.25V, 5mA
50
t (ns)

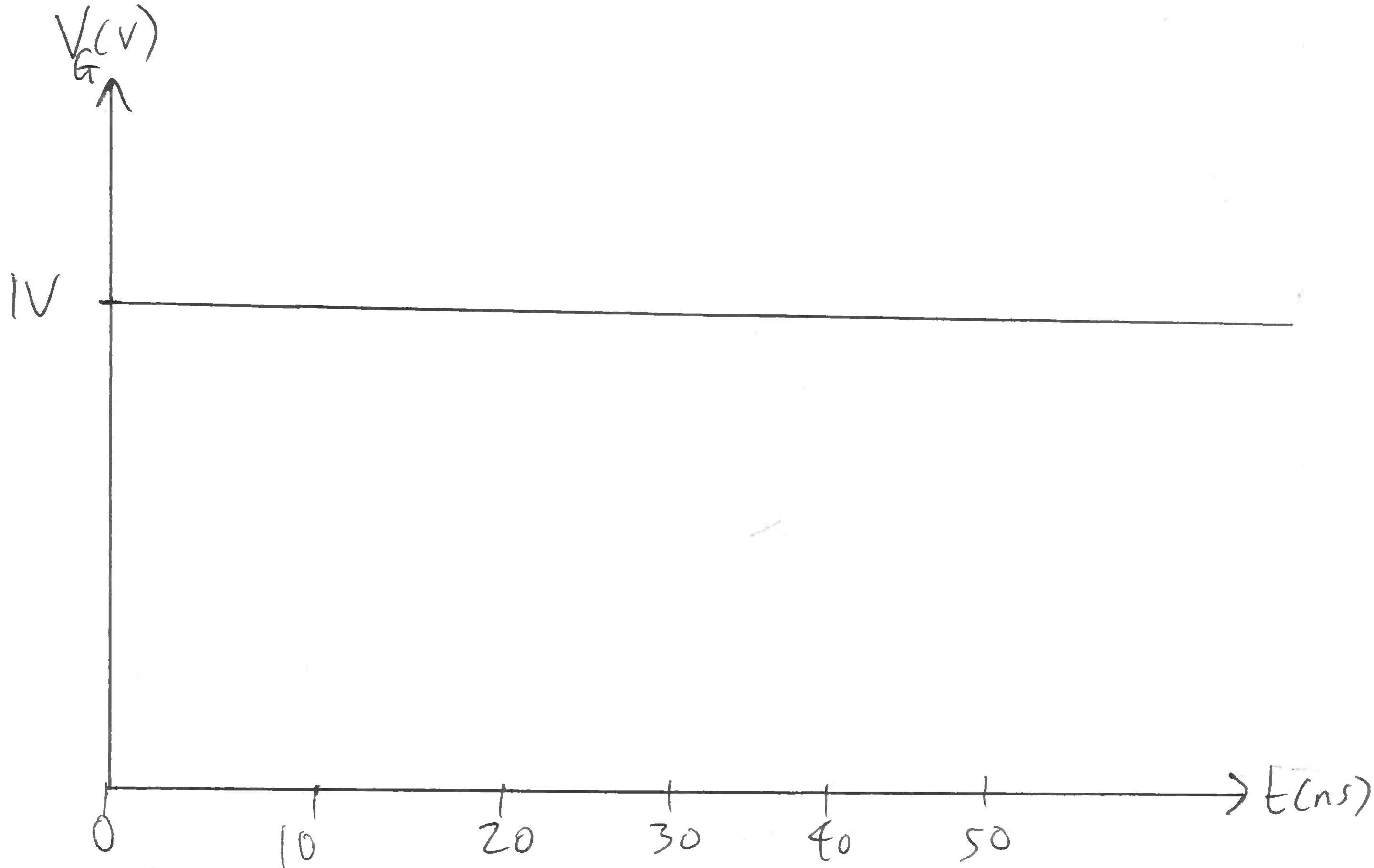


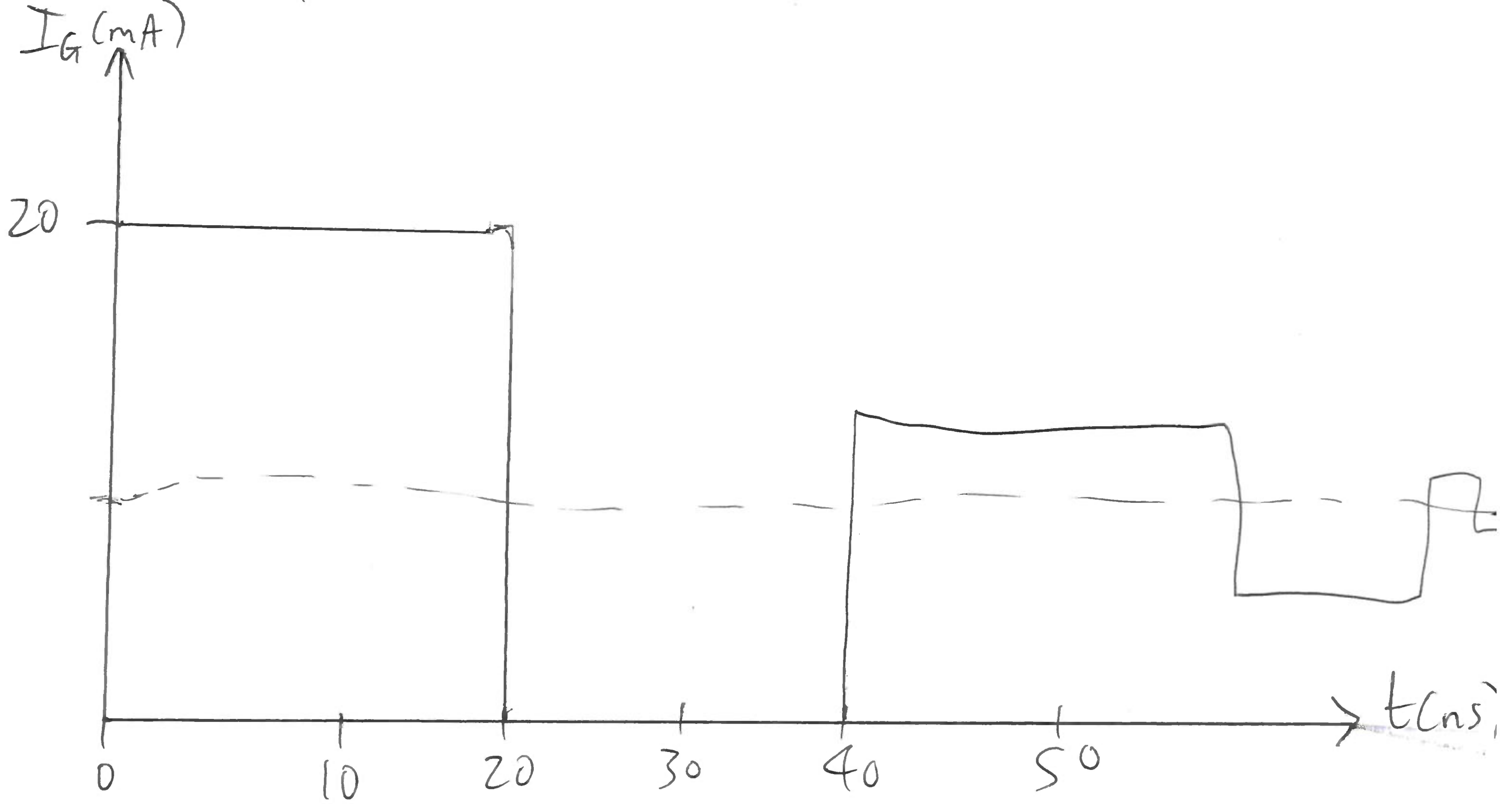
V_L (V)
1.5
1
0.75
0
10
20
30
40
50
t (ns)



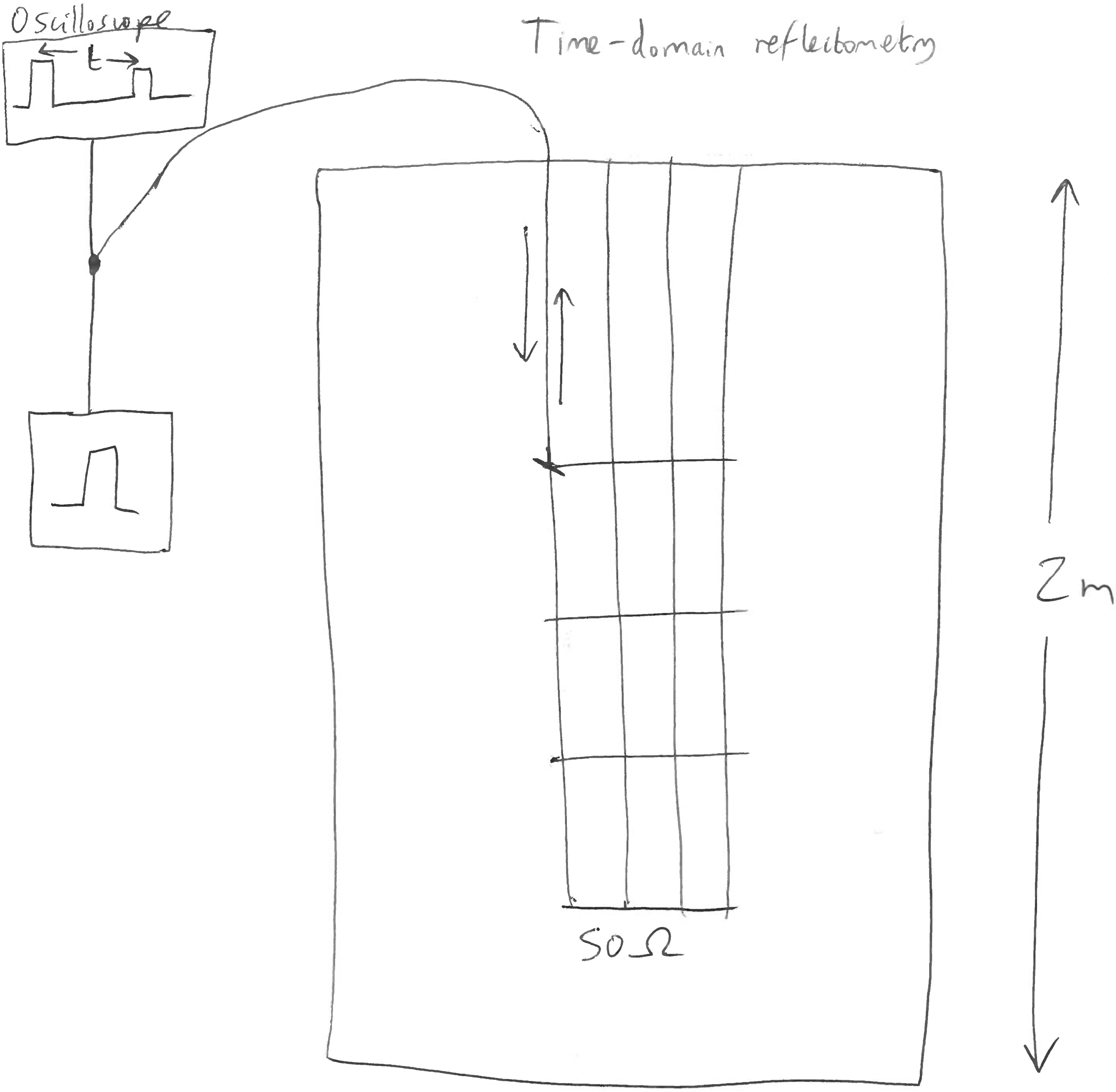
$$I_L(t \to \infty) = \frac{1V}{150\Omega} = 6.67mA$$

$$V = IR \Rightarrow R_L = \frac{V_L}{I_L} = 150\Omega$$





$t = \infty$: $I_G \rightarrow I_L = 6.67\,mA$



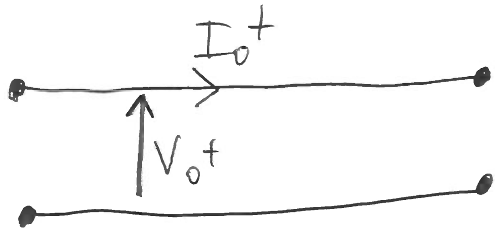
$$d = v_p t = 2\times10^{8} \times 5\times10^{-9} = 1\,m$$

1) Lumped impedance:

$$Z = \frac{V}{I}$$

Can measure with DC multimeter

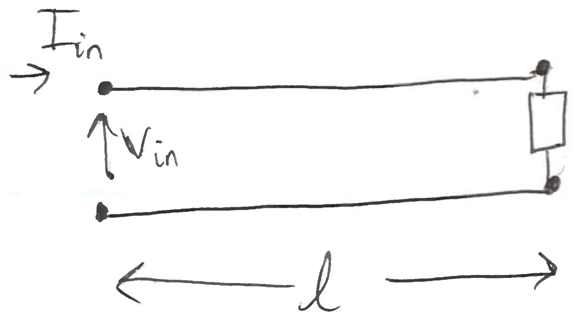
2) Characteristic impedance:



$$Z_0 = \frac{V_0^+}{I_0^+}$$

Can't measure with multimeter

3) Input impedance:



$$Z_{in} = \frac{V_{in}}{I_{in}}$$

Can measure with AC multimeter

e.g.